Predicting Board Game Ratings Using Machine Learning

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Course: Artificial Intelligence

1 Introduction

This project focuses on developing a predictive modeling pipeline to analyze and predict board game ratings using structured metadata from the BoardGameGeek dataset. A data-centric approach is followed, beginning with raw data acquisition, followed by cleaning, feature engineering, and exploratory data analysis (EDA). The goal is to uncover meaningful patterns that influence board game success and provide a strong foundation for predictive modeling.

2 Problem Definition

2.1 Problem Statement

The rapid growth of the board game industry has resulted in a vast number of games with diverse characteristics such as mechanics, complexity, playtime, and player count. However, understanding which of these features contribute to higher user ratings remains a challenge. This project aims to address this gap by developing a predictive framework that estimates board game ratings based on these attributes.

2.2 Target Audience

The primary audience includes board game designers, publishers, and data analysts interested in understanding the factors influencing game success. Additionally, this work can support the development of recommendation systems and data-driven decision-making in the gaming industry.

2.3 AI Justification

The relationship between board game features and user ratings is inherently complex and involves multiple interacting variables. Traditional rule-based systems are insufficient to capture such non-linear relationships. Machine learning techniques can learn hidden patterns from large datasets and generalize effectively, making them well-suited for this predictive task.

3 Dataset Description

The dataset used in this project is sourced from BoardGameGeek and obtained via Kaggle. It contains detailed information about thousands of board games, including attributes such as player count, average playtime, and complexity. The complexity of a game is represented using the BoardGameGeek "weight" metric, which reflects the perceived difficulty of learning and playing the game.

4 Data Acquisition

The dataset was collected from Kaggle as raw, unprocessed data. It contains missing values, duplicate entries, and categorical variables that require preprocessing before being used for analysis and modeling.

5 Data Cleaning

5.1 Handling Missing Values

Missing values were identified and handled using appropriate techniques such as imputation or removal, depending on the relevance of the feature. This ensured that the dataset remained consistent and usable for further analysis.

5.2 Duplicate Removal

Duplicate records were removed to eliminate redundancy and prevent bias in downstream analysis.

5.3 Outlier Handling

Outliers in numerical features such as playtime and ratings were analyzed and managed to improve data quality and ensure more reliable statistical analysis.

6 Feature Engineering

6.1 Feature Selection

Relevant features such as player count, playtime, and complexity were retained, while irrelevant or redundant features were removed to improve model efficiency.

6.2 Encoding and Transformation

Categorical variables, including game mechanics, were converted into numerical representations using encoding techniques. Numerical features were scaled and normalized to ensure uniformity across the dataset.

7 Exploratory Data Analysis

7.1 Distribution Analysis

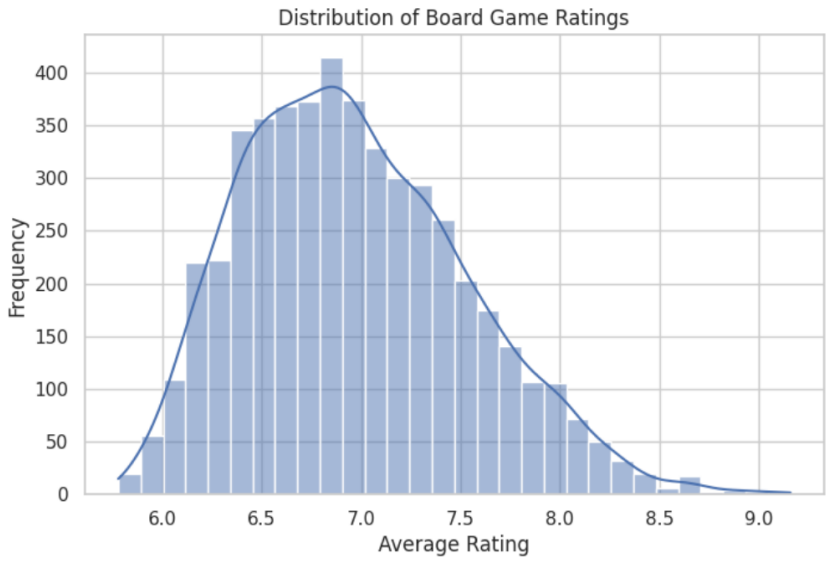

Figure 1: Distribution of board game ratings showing concentration around moderate to high values.

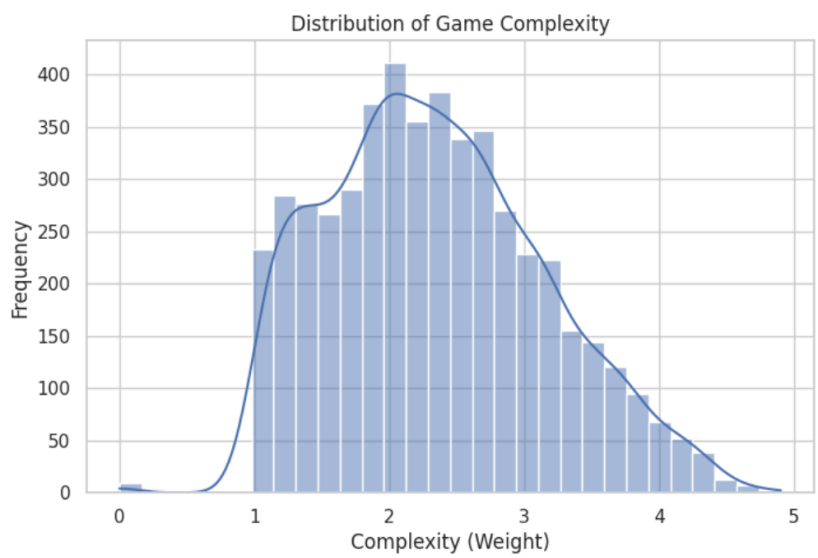

Figure 2: Distribution of game complexity (weight), indicating a wide range of difficulty levels.

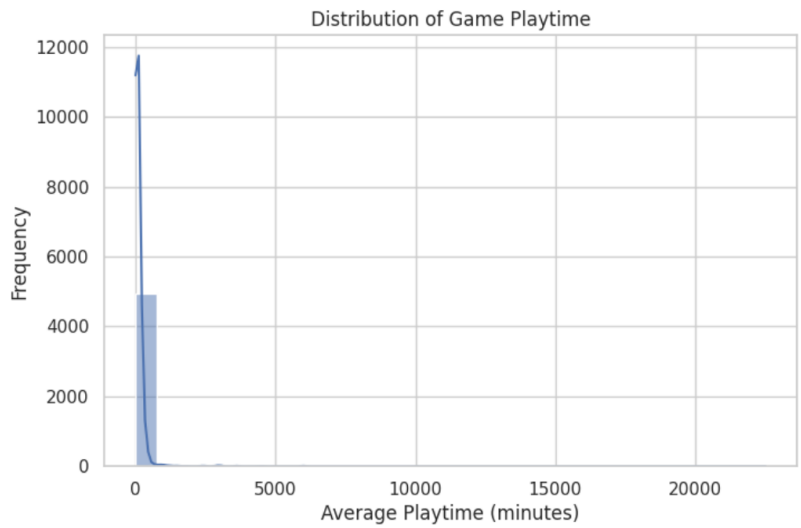

Figure 3: Distribution of average playtime, showing variability in game durations.

The distribution analysis indicates that most board games receive moderate to high ratings, suggesting a general positive bias among users. Complexity and playtime distributions reveal a wide diversity in game design, ranging from simple and short games to highly complex and lengthy ones.

7.2 Correlation Analysis

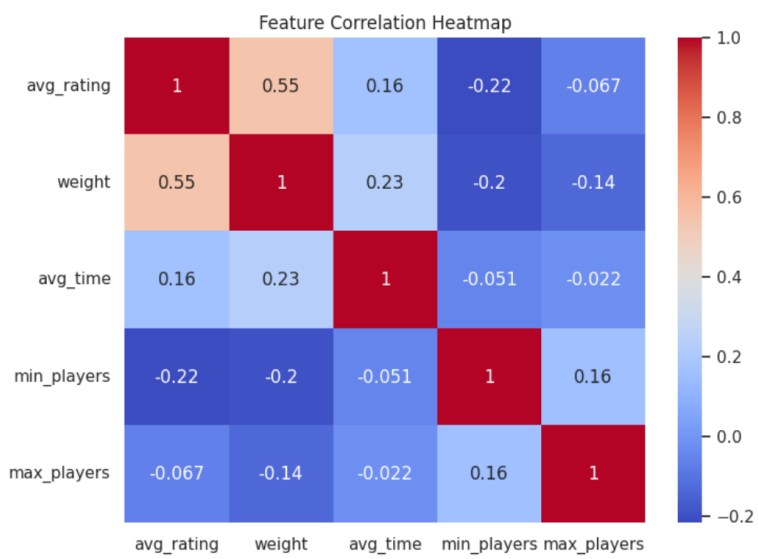

Figure 4: Correlation heatmap showing relationships between key numerical features.

The correlation analysis highlights weak to moderate relationships between features such as complexity, playtime, and ratings. This suggests that no single feature strongly determines a game's rating, reinforcing the need for machine learning models to capture combined effects.

7.3 Feature Impact Exploration

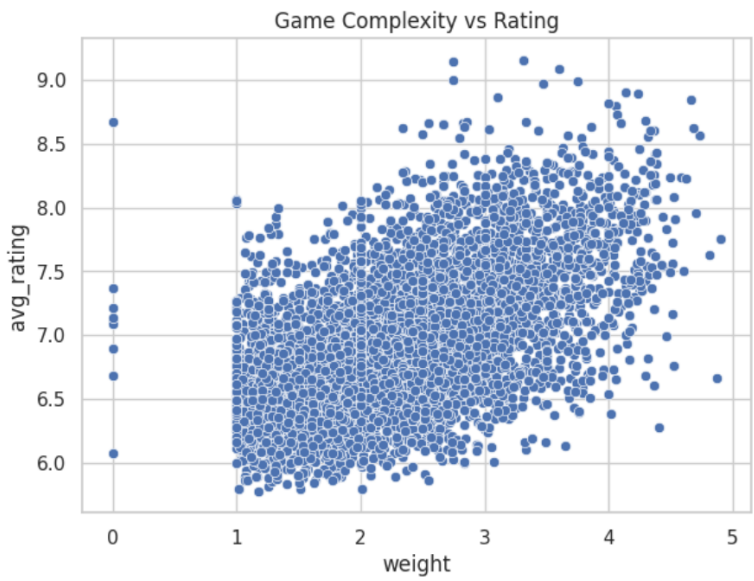

Figure 5: Relationship between game complexity and ratings.

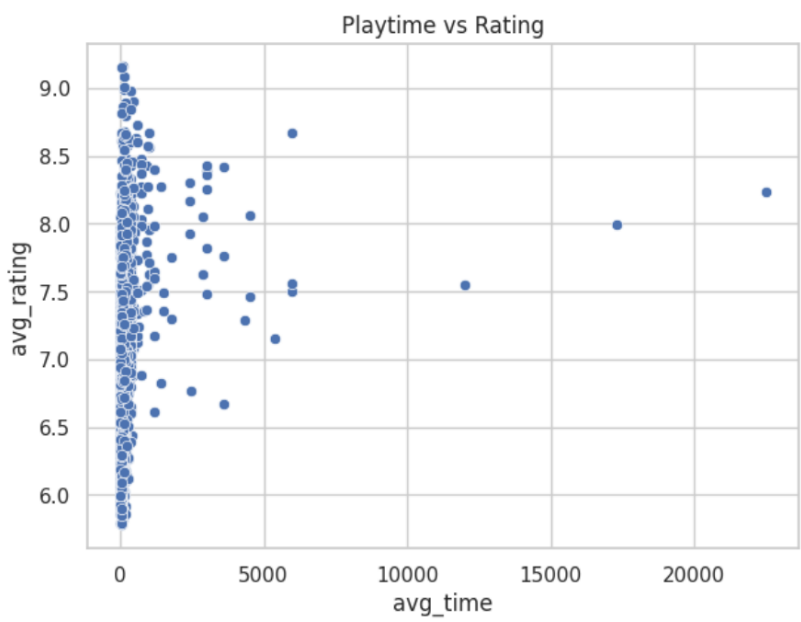

Figure 6: Relationship between playtime and ratings.

The feature impact analysis suggests that moderately complex games tend to achieve slightly higher ratings, while extremely simple or highly complex games exhibit greater variability. Similarly, playtime shows no strong linear relationship with ratings, indicating that user preferences vary across different game durations.

8 Conclusion

This report presented the initial phases of the AI project, including problem definition, data acquisition, preprocessing, and exploratory data analysis. The results demonstrate that board game ratings are influenced by multiple interacting factors rather than a single dominant feature. These insights provide a strong foundation for developing and evaluating predictive machine learning models in the next phase of the project.